

IEOR 165 Course Project

House Price Prediction via Linear, Regularized, and Kernel Regression

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1 Introduction

The price of a house depends on a number of measurable factors — floor area, the number of bedrooms and bathrooms, furnishings, proximity to a main road, and so on. The goal of this project is to quantify these effects and predict house prices from them. Three regression models are fit on the same training data and compared on a held-out test set:

- (1) Ordinary Least Squares (OLS),
- (2) Regularized regression (LASSO), with the penalty λ selected by 5-fold cross-validation, and
- (3) Nadaraya–Watson kernel regression on a single feature, with the bandwidth h selected by 5-fold cross-validation.

For models (2) and (3) we report the cross-validation error as a function of the tuning parameter and the value selected. For all three models, we report a closed-form prediction rule $\hat{y} = f(x)$ and the test-set mean squared error (MSE).

The models shown will not have any performance-increasing methods done to them, i.e feature-selection, $\log(\text{price})$, etc.

2 Data and Preprocessing

The dataset (`house_sales.csv`) contains 545 observations on house price together with 12 features — 5 numeric (`area`, `bedrooms`, `bathrooms`, `stories`, `parking`) and 7 categorical (`mainroad`, `guestroom`, `basement`, `hotwaterheating`, `airconditioning`, `prefarea`, `furnishingstatus`). Categorical features were one-hot encoded with `drop_first=True`, yielding 13 predictors plus an intercept.

The data was split 80/20 into training ($n_{\text{train}} = 436$) and test ($n_{\text{test}} = 109$) sets using `random_state=42`. For the LASSO model, the training features were additionally standardized (zero mean, unit variance), with the same scaling applied to the test set.

3 Model 1 — Ordinary Least Squares

Formula

With the 13 encoded predictors, the fitted OLS model is

$$\begin{aligned}\hat{y} = & 2.600 \times 10^5 + 235.969 (\text{area}) + 7.678 \times 10^4 (\text{bedrooms}) + 1.094 \times 10^6 (\text{bathrooms}) \\ & + 4.075 \times 10^5 (\text{stories}) + 2.248 \times 10^5 (\text{parking}) + 3.679 \times 10^5 (\text{mainroad_yes}) \\ & + 2.316 \times 10^5 (\text{guestroom_yes}) + 3.903 \times 10^5 (\text{basement_yes}) + 6.846 \times 10^5 (\text{hotwaterheating_yes}) \\ & + 7.914 \times 10^5 (\text{airconditioning_yes}) + 6.299 \times 10^5 (\text{prefarea_yes}) \\ & - 1.269 \times 10^5 (\text{semi-furnished}) - 4.136 \times 10^5 (\text{unfurnished}).\end{aligned}$$

Fit on the training set

The training fit gives $R^2 = 0.686$ and adjusted $R^2 = 0.676$, with an overall F -statistic of 70.90 ($p \approx 2.16 \times 10^{-97}$). Coefficients on `area`, `bathrooms`, `stories`, `parking`, `mainroad_yes`, `basement_yes`, `hotwaterheating_yes`, `airconditioning_yes`, and `prefarea_yes` are all significant at the 5% level. The `bedrooms`, `guestroom_yes`, and `furnishingstatus_semi-furnished` coefficients are not significant ($p = 0.309$, 0.100 , and 0.299 respectively).

Table 1: OLS coefficient estimates (training set, $n = 436$).

Variable	Coefficient	Std. Err.	t	$P > t $
const	2.600e+05	2.74e+05	0.949	0.343
area	2.360e+02	2.49e+01	9.476	0.000
bedrooms	7.678e+04	7.54e+04	1.018	0.309
bathrooms	1.094e+06	1.14e+05	9.564	0.000
stories	4.075e+05	6.88e+04	5.919	0.000
parking	2.248e+05	6.18e+04	3.639	0.000
mainroad_yes	3.679e+05	1.49e+05	2.464	0.014
guestroom_yes	2.316e+05	1.41e+05	1.648	0.100
basement_yes	3.903e+05	1.18e+05	3.320	0.001
hotwaterheating_yes	6.846e+05	2.23e+05	3.070	0.002
airconditioning_yes	7.914e+05	1.17e+05	6.772	0.000
prefarea_yes	6.299e+05	1.21e+05	5.205	0.000
furnishingstatus_semi-furnished	-1.269e+05	1.22e+05	-1.041	0.299
furnishingstatus_unfurnished	-4.136e+05	1.34e+05	-3.077	0.002

Test-set performance

- Test $R^2 = 0.6529$
- Test MSE = 1,754,318,687,330
- Test RMSE = 1,324,507

Note that the Test MSE is a very high number, which can be attributed to the prices of the house being in the millions of dollars, so when we find the MSE (Mean Squared Error) between millions of dollars, it will be $1,000,000^2$, which comes to one trillion. A more interpretable number is the RMSE (Root Mean Squared Error).

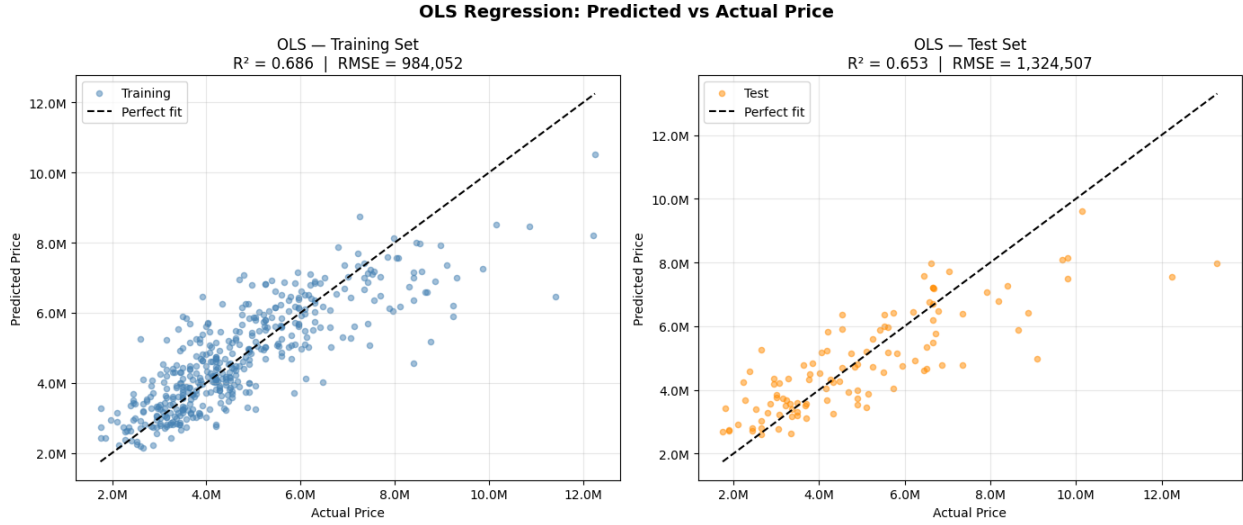


Figure 1: OLS predicted vs. actual prices on the training (left) and test (right) sets. The dashed line is the perfect-prediction reference. The model generalizes well: the training $R^2 = 0.686$ and test $R^2 = 0.653$ are close, indicating no severe overfitting.

4 Model 2 — LASSO Regression

LASSO solves the penalized least-squares problem

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1.$$

The features were standardized prior to fitting (regularization is scale-sensitive); the response was kept on its original scale.

Selecting λ via 5-fold cross-validation

We searched 200 candidate values of λ on a logarithmic grid from 10^4 to 10^9 . For each value we ran 5-fold cross-validation on the training set and recorded the mean MSE.

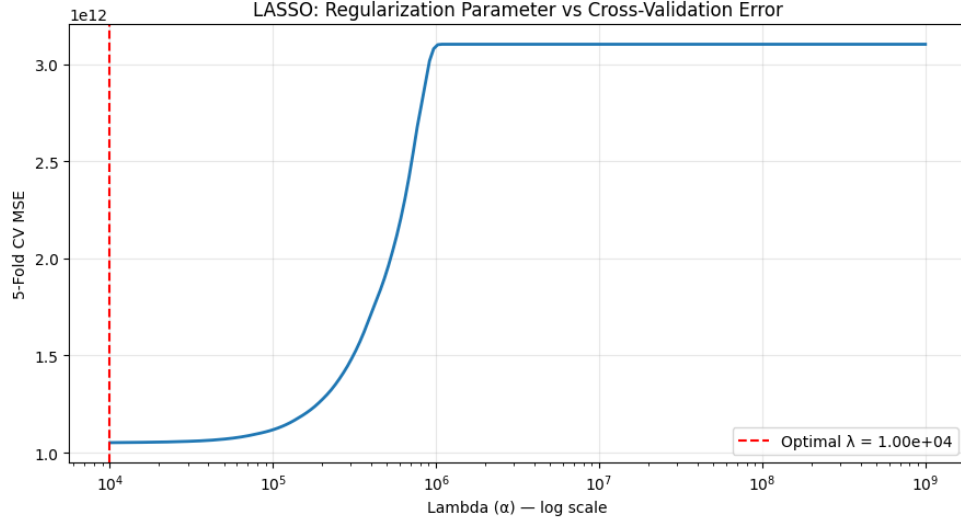


Figure 2: LASSO 5-fold cross-validation MSE as a function of λ (log scale). The CV error is essentially flat for small λ and rises sharply once the penalty becomes large enough to shrink coefficients toward zero. The dashed red line marks the minimizing value $\lambda^* = 1.000 \times 10^4$.

Formula

With the optimal $\lambda^* = 1.000 \times 10^4$, the fitted LASSO model (in standardized feature units) is

$$\begin{aligned} \hat{y} = & 4,706,527.39 + 517,428.79 (\text{area}_{\text{std}}) + 54,260.41 (\text{bedrooms}_{\text{std}}) + 519,105.43 (\text{bathrooms}_{\text{std}}) \\ & + 344,636.10 (\text{stories}_{\text{std}}) + 188,927.57 (\text{parking}_{\text{std}}) + 124,881.40 (\text{mainroad_yes}_{\text{std}}) \\ & + 85,024.33 (\text{guestroom_yes}_{\text{std}}) + 181,523.73 (\text{basement_yes}_{\text{std}}) + 139,644.56 (\text{hotwaterheating_yes}_{\text{std}}) \\ & + 362,612.26 (\text{airconditioning_yes}_{\text{std}}) + 261,253.97 (\text{prefarea_yes}_{\text{std}}) \\ & - 41,419.60 (\text{semi-furnished}_{\text{std}}) - 174,168.80 (\text{unfurnished}_{\text{std}}), \end{aligned}$$

where each $x_{\text{std}} = (x - \bar{x}_{\text{train}})/s_{\text{train}}$.

Table 2: LASSO coefficients on standardized features ($\lambda = 10^4$). At this value, the penalty is light enough that no feature is driven exactly to zero — LASSO behaves like a slightly shrunk OLS.

Feature	Coefficient
intercept	4,706,527.39
area	517,428.79
bedrooms	54,260.41
bathrooms	519,105.43
stories	344,636.10
parking	188,927.57
mainroad_yes	124,881.40
guestroom_yes	85,024.33
basement_yes	181,523.73
hotwaterheating_yes	139,644.56
airconditioning_yes	362,612.26
prefarea_yes	261,253.97
furnishingstatus_semi-furnished	-41,419.60
furnishingstatus_unfurnished	-174,168.80

Test-set performance

- Test $R^2 = 0.6502$
- Test MSE = 1,768,239,121,769
- Test RMSE = 1,329,752

LASSO performs almost identically to OLS on this dataset: at the chosen λ^* no coefficient is zeroed out, and the test MSE differs from OLS by only about 0.8%.

5 Model 3 — Nadaraya–Watson Kernel Regression

We use Nadaraya–Watson on the single feature most strongly associated with price: **area**. With a Gaussian kernel $K(u) = \exp(-u^2/2)$ and bandwidth h , the prediction at a query point x^* is

$$\hat{y}(x^*) = \frac{\sum_{i=1}^n K\left(\frac{x^* - x_i}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{x^* - x_i}{h}\right)},$$

where $\{(x_i, y_i)\}_{i=1}^n$ are the training pairs of (area, price).

Selecting the bandwidth h via 5-fold cross-validation

We searched 150 candidate bandwidths on a logarithmic grid from 10^2 to $10^{4.5}$ (in the original units of **area**, square feet). For each h , we ran 5-fold cross-validation on the training set and recorded the mean MSE.

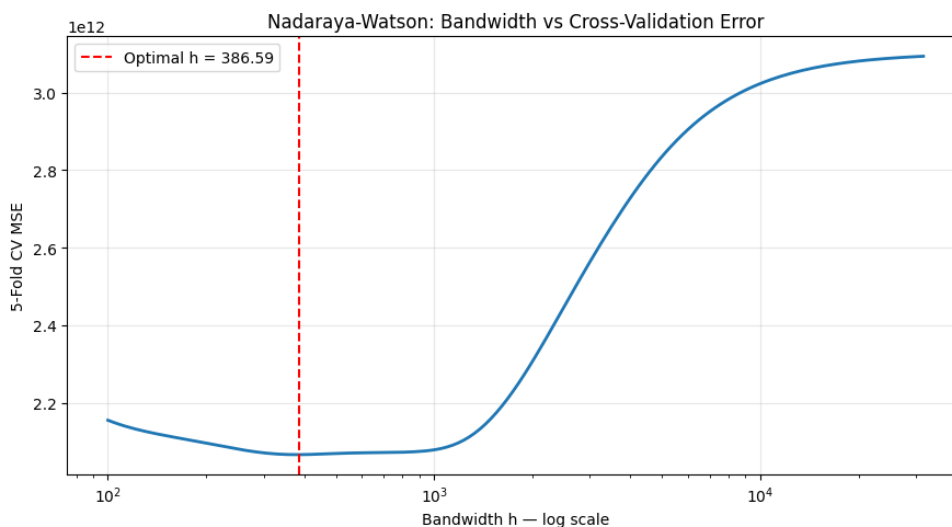


Figure 3: Nadaraya–Watson 5-fold cross-validation MSE as a function of bandwidth h (log scale). The curve has the classic U-shape: very small bandwidths overfit (high variance), very large bandwidths oversmooth (high bias toward the global mean). The optimum is at $h^* = 386.59$.

Formula

With $h^* = 386.59$, the prediction rule becomes

$$\hat{y}(x^*) = \frac{\sum_{i=1}^{436} \exp\left(-\frac{1}{2} \left(\frac{x^* - x_i}{386.59}\right)^2\right) y_i}{\sum_{i=1}^{436} \exp\left(-\frac{1}{2} \left(\frac{x^* - x_i}{386.59}\right)^2\right)}$$

where x^* is the area (sq ft) of the house being predicted and $\{(x_i, y_i)\}$ are the 436 training observations.

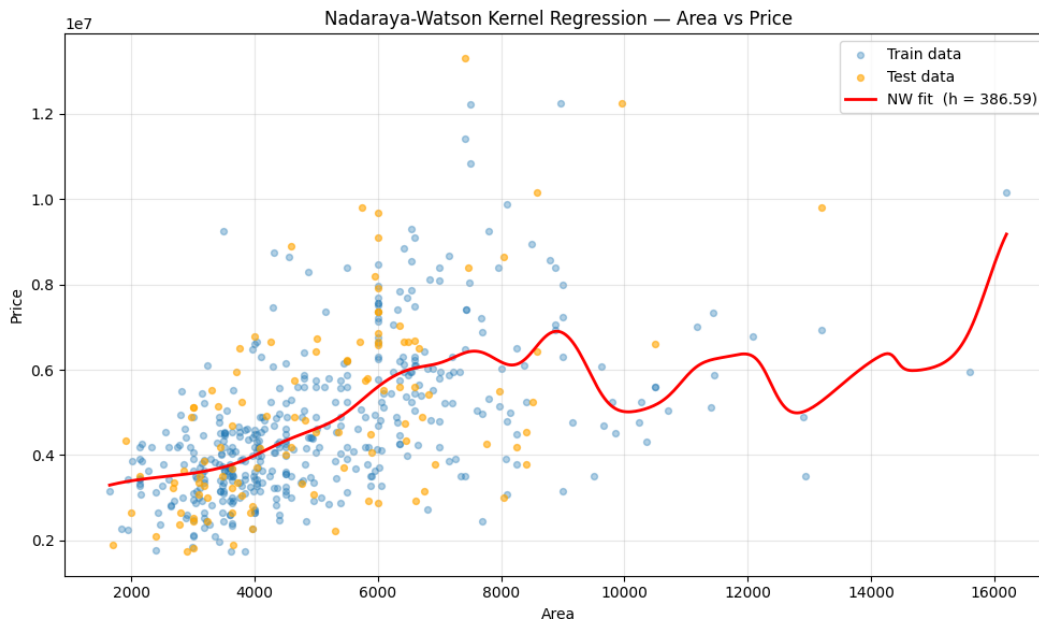


Figure 4: Nadaraya–Watson fitted curve over the training and test data. The fit captures a clear, gently nonlinear increase in price with area, but a single feature plainly cannot explain the large vertical spread, which is driven by the omitted features (bathrooms, AC, location, etc.).

Test-set performance

- Test $R^2 = 0.2475$
- Test MSE = 3,803,754,026,232
- Test RMSE = 1,950,322

6 Comparison and Discussion

Table 3: Test-set performance summary.

Model	Test MSE	Test RMSE	Test R^2
OLS	1,754,318,687,330	1,324,507	0.6529
LASSO ($\lambda = 10^4$)	1,768,239,121,769	1,329,752	0.6502
Nadaraya–Watson ($h = 386.59$)	3,803,754,026,232	1,950,322	0.2475

OLS and LASSO produce essentially identical predictive performance. At the cross-validated $\lambda^* = 10^4$ the LASSO penalty is light enough that none of the coefficients is driven to zero, so the LASSO solution is a slightly shrunk OLS solution. The training-set fit ($R^2 = 0.686$) and the test-set fit ($R^2 \approx 0.65$) are close, so there is no evidence of harmful overfitting that LASSO would need to correct.

Nadaraya–Watson is markedly worse, with test MSE roughly $2.2\times$ that of OLS. This is expected: the kernel regressor uses only **area**, while OLS and LASSO use all 13 features. The remaining $\sim 75\%$ of the variance in price simply cannot be explained by **area** alone, regardless of how flexibly the regression curve is shaped. The takeaway is that for this dataset the gain from including additional features (bathrooms, AC, preferred area, etc.) is far larger than any gain from modeling **area** nonlinearly.

Recommended model. OLS is the model of choice here: it has the lowest test MSE, an interpretable closed-form prediction rule, and standard inference for the coefficients.

Source code

The full implementation (data loading, model fitting, cross-validation, and plotting) is available at: <https://github.com/chwu7/165-Project/tree/main>.